

MATHEMATICAL PROMPTS FOR A 7-DIMENSIONAL UNIVERSE

Developed as part of a theoretical framework exploring extended dimensions in cosmology and physics

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Abstract

This document contains three mathematical prompts designed for computational exploration of a 7-dimensional universe. These prompts were developed collaboratively using advanced large language models (GPT-3.5, GPT-4, and Gemini) and form the computational foundation for modified Einstein field equations integrating the dimensions of zero, infinity, and chance.

The prompts are optimized for use in theoretical physics, cosmological modeling, and quantum mechanics frameworks. These prompts are part of the supporting work, “Unraveling the Mysteries of a 7-Dimensional Universe: A Modified Hypothesis of General Relativity,” which provides the theoretical and mathematical context for their creation.

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Original Prompt GPT (2023)

In our 7-dimensional universe, space-time is described by a metric tensor with seven components, including the additional dimensions of zero, infinity, and chance. This is represented by the metric tensor $g_{\{\mu\nu\}}$ with the components:

$$g_{00} = -c^2 + \frac{2GM}{r} - \frac{H^2 r^2}{3} + \frac{C\xi}{\theta}$$

$$g_{11} = \frac{1}{1 - \frac{2GM}{r}}$$

$$g_{22} = r^2, \quad g_{33} = r^2 \sin^2 \theta$$

$$g_{44} = \frac{1}{\infty}, \quad g_{55} = \infty, \quad g_{66} = \xi^2$$

Using this metric tensor, we can derive the Christoffel symbols and Riemann curvature tensor to study the curvature of space-time. The Christoffel symbols are given by:

$$\Gamma^\mu_{\nu\rho} = \frac{1}{2} g^{\mu\sigma} \left(\partial_\nu g_{\rho\sigma} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho} \right)$$

The Riemann curvature tensor is given by:

$$R^\mu_{\nu\sigma\rho} = \partial_\sigma \Gamma^\mu_{\nu\rho} - \partial_\rho \Gamma^\mu_{\nu\sigma} + \Gamma^\mu_{\alpha\sigma} \Gamma^\alpha_{\nu\rho} - \Gamma^\mu_{\alpha\rho} \Gamma^\alpha_{\nu\sigma}$$

The Ricci curvature tensor and scalar curvature are calculated from the Riemann curvature tensor as follows:

$$R_{\mu\nu} = R^{\alpha}{}_{\mu\alpha\nu}, \quad R = g^{\mu\nu} R_{\mu\nu}$$

The Einstein tensor is given by:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu}$$

Where Λ is the cosmological constant.

Gemini Optimized Prompt (2024)

<<Task: Calculate the Einstein tensor for a 7-dimensional universe (7dU) with the following metric tensor:>>

$$g_{00} = -c^2 + \frac{2GM}{r} - \frac{H^2 r^2}{3} + \frac{C\xi}{\theta}$$

$$g_{11} = \frac{1}{1 - \frac{2GM}{r}}$$

$$g_{22} = r^2, \quad g_{33} = r^2 \sin^2 \theta$$

$$g_{44} = \frac{1}{\infty}, \quad g_{55} = \infty, \quad g_{66} = \xi^2$$

Use the following formulas:

- Christoffel symbols:

$$\Gamma^\mu_{\nu\rho} = \frac{1}{2} g^{\mu\sigma} \left(\partial_\nu g_{\rho\sigma} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho} \right)$$

- Riemann curvature tensor:

$$R^{\mu}{}_{\nu\sigma\rho} = \partial_{\sigma}\Gamma^{\mu}{}_{\nu\rho} - \partial_{\rho}\Gamma^{\mu}{}_{\nu\sigma} + \Gamma^{\mu}{}_{\alpha\sigma}\Gamma^{\alpha}{}_{\nu\rho} - \Gamma^{\mu}{}_{\alpha\rho}\Gamma^{\alpha}{}_{\nu\sigma}$$

- Ricci curvature tensor:

$$R_{\mu\nu} = R^{\alpha}{}_{\mu\alpha\nu}$$

- Scalar curvature:

$$R = g^{\mu\nu}R_{\mu\nu}$$

- Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu}$$

Where Λ is the cosmological constant.

GPT Optimized Prompt (2024)

<<Task: Calculate the Einstein tensor $G_{\mu\nu}$ for a 7-dimensional universe (7dU) using the following metric tensor components:>>

$$g_{00} = -c^2 + \frac{2GM}{r} - \frac{H^2 r^2}{3} + \frac{C\xi}{\theta}, \quad g_{11} = \frac{1}{1 - \frac{2GM}{r}}$$

$$g_{22} = r^2, \quad g_{33} = r^2 \sin^2 \theta$$

$$g_{44} = \frac{1}{\infty}, \quad g_{55} = \infty, \quad g_{66} = \xi^2$$

Using the following formulas:

- Christoffel symbols:

$$\Gamma^\mu_{\nu\rho} = \frac{1}{2} g^{\mu\sigma} \left(\partial_\nu g_{\rho\sigma} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho} \right)$$

- Riemann curvature tensor:

$$R^\mu_{\nu\sigma\rho} = \partial_\sigma \Gamma^\mu_{\nu\rho} - \partial_\rho \Gamma^\mu_{\nu\sigma} + \Gamma^\mu_{\alpha\sigma} \Gamma^\alpha_{\nu\rho} - \Gamma^\mu_{\alpha\rho} \Gamma^\alpha_{\nu\sigma}$$

- Ricci curvature tensor:

$$R_{\mu\nu} = R^{\alpha}_{\mu\alpha\nu}$$

- Scalar curvature:

$$R = g^{\mu\nu} R_{\mu\nu}$$

- Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu}$$

Where Λ is the cosmological constant.